

Practice Questions

Test your comprehension by answering the following questions:

1. What is $4!$ (that is, the factorial of 4)?
2. How can you use the factorial of $(n - 1)$ to calculate the factorial of n ?
3. What is the critical weakness of the recursive factorial function?
4. What are the first five numbers of the Fibonacci sequence?
5. What two numbers do you add to get the n th Fibonacci number?
6. What is the critical weakness of the recursive Fibonacci function?
7. What does an iterative algorithm always use?
8. Is it always possible to convert an iterative algorithm into a recursive one?
9. Is it always possible to convert a recursive algorithm into an iterative one?
10. Any recursive algorithm can be performed iteratively by using what two things?
11. What three features do programming problems that are suitable to recursive solutions have?
12. When is recursion required to solve a programming problem?